

the task. A shuttle would take off from the MHF and supply inbound and outbound resources to the AFC, as shown in Fig. 1(b).

The AFC delivery system could let customers receive their products within ten minutes of placing the order. The AFC would maintain some inventory so that a UAV can leave with the ordered goods immediately after receipt of the order. The UAV deployed from the AFC would quickly descend towards the customer to deliver the items. During the descent, the UAV would use little or no power. Hence, Amazon would be able to provide faster services to customers located far away and its UAVs would safely return to MHF after delivery.

One of Amazon's primary objectives is to secure and expand its ecosystem by attracting more sellers and customers to its marketplace. A recent study showed that nearly 25% people are willing to pay a substantial premium for same-day or immediate delivery. The rapid delivery capability of the AFC system can enlarge Amazon's ecosystem by increasing the number of Amazon Prime subscribers and sellers.

Shipping cost is also a major concern for Amazon, as it incurs a net loss on shipping annually. In this regard, the AFC delivery system can reduce transportation costs and operating energy by staying or moving in the stratosphere where air density and resistance are low. The AFC will use electric and renewable energy such as solar energy. These factors will enable the company to provide a sustainable membership service by reducing shipping costs

American e-commerce giant Amazon now has patents for flying drone warehouse, an underwater warehouse, a bee-hive like warehouse, and mobile ground based fulfilment centres. As per the latest reports from CB Insights, the retail giant has been focussed on shortening the distance between warehouses and consumers for some time. It has considered various formats for warehouse network expansions like flying warehouse, mobile truck mini warehouse, underwater warehouse, local re-stocking stations for drones, and multi-level drone-docking fulfilment centre.

Use of UAVs in industries

UAVs present a powerful tool for collecting data remotely. In inspection scenarios, using a UAV to collect data instead of a person doing so can make a big impact on safety, since it reduces the exposure of personnel to potentially dangerous scenarios such as climbing a cell tower or walking along scaffolding inside a giant tank to collect visual data. People also use UAVs to collect data for surveying, mapping, or even to help investigators find human remains.

Here are the primary types of data currently collected by drones:

Visual data. This is by far the most common type of data a drone is used to collect. By flying over an area or object of interest, a drone can help see things that might not be otherwise visible and collect a record of what is seen.

Thermal data. After visual data, thermal data is one of the most common types of data industries collect using drones. Aerial thermal data can help firefighters determine where to focus their efforts during a raging fire, or help inspectors identify potential problem areas in a solar array.

Lidar data. A light detection and ranging (lidar) sensor 'illuminates' a target with a laser light and measures the reflection to create data points that can be used to map the area. Aerial lidar can be used to help companies in various industries to create 3D maps of an area, which can be used for project planning or progress tracking. Since lidar can penetrate tree cover and even earth to reveal structures hidden underground, it has also been used by archaeologists to help them discover new sites of interest for excavation.

Multispectral data. Multispectral data is collected by sensors that measure reflected energy within several specific sections (or bands) of the electromagnetic spectrum. Aerial multispectral data can be used in agriculture and conservation to monitor plant and tree health, and it is also being used by law enforcement agencies to help find human remains.

Hyperspectral data. Hyperspectral sensors measure energy in narrower and more numerous bands than multispectral sensors. Aerial hyperspectral data can be used in agriculture for

monitoring the health of crops, and in security and defence for detecting the presence of those who should not be in a given area.

Here are some of the most common deliverables created from drone data:

Photos and videos. One of the most common deliverables for commercial drone work are still photographs and videos. These might be for professional photography or videography purposes (at weddings or family photo shoots), aerial shots of real estate to help market it, or even high-end cinematography for filmmaking.

3D maps. Across various industries 3D maps are becoming a common deliverable for drone data, helping people to better visualise the spaces in which they are working.

Orthomosaics. An orthomosaic is a photo representation of an area created by stitching together several photos. Orthomosaics are used in construction to visualise building sites, in public safety to record the details of places where large groups of people commonly gather, or in civil engineering to track the progress of a large project such as restoring part of a beach.

Actionable reports. In some industries a deliverable produced from drone data could include a report generated by industry-specific software. For example, Pix4D's agriculture-specific software Pix4Dfields allows users to produce agricultural indices to better understand plant stress as well as aggregating vegetation index maps into zones.

Following are some of the top industries and sectors using drones for commercial purposes today, along with the ways that they are using them:

Agriculture. Farmers use drones to collect data on their crops and then use that data to improve their yields.

Chemicals. Drones are being used in the chemicals industry to improve indoor inspections by taking the place of inspectors in collecting visual data inside large assets used in chemical processes.

Conservation. One of the main ways drones are helping conservation efforts is by providing detailed vegetation maps to help track forestry work and water mapping to better understand how water moves through an area. Drones that shoot out